Skill Week-7

Session-13

Task 1: To parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

```
kiran@k:~/OSSP$ cd ~/OSSP
rm -f input_redirect.c input_redirect
nano 13_task1.c
kiran@k:~/OSSP$ gcc 13_task1.c -o task1
kiran@k:~/OSSP$ printf "Line 1\nLine 2\nLine 3\nLine 4\n" > input
.txt
kiran@k:~/OSSP$ ./task1
Session 11 - Input Redirection
Enter command with input redirection:
myshell> wc < input.txt

Command           : wc
Input file        : input.txt
Redirection       : stdin < file

Input file opened successfully.
Executing command with redirected stdin...
      4       8      28

Input redirection completed successfully.
File descriptor cleaned up.
Parent process restored normal stdin.
kiran@k:~/OSSP$
```

kiran@k: ~/OSSP

GNU nano 8.7.1 13_task1.c

```c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char input[100];
    char command[50];
    char filename[50];

    printf("Session 11 - Input Redirection\n");
    printf("Enter command with input redirection:\n");

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    if (sscanf(input, "%s < %s", command, filename) != 2)
    {
        printf("Invalid input redirection format.\n");
        return 1;
    }

    printf("\nCommand           : %s\n", command);
    printf("Input file        : %s\n", filename);
    printf("Redirection       : stdin < file\n\n");

    int fd = open(filename, O_RDONLY);
```

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GNU nano 8.7.1 13_task1.c

```c
    int fd = open(filename, O_RDONLY);

    if (fd < 0)
    {
        perror("Input file");
        return 1;
    }

    printf("Input file opened successfully.\n");
    printf("Executing command with redirected stdin...\n");

    pid_t pid = fork();

    if (pid == 0)
    {
        dup2(fd, STDIN_FILENO);
        close(fd);

        execlp(command, command, NULL);

        perror("exec");
        exit(1);
    }

    close(fd);
    wait(NULL);

    printf("\nInput redirection completed successfully.\n");
    printf("File descriptor cleaned up.\n");
    printf("Parent process restored normal stdin.\n");

    return 0;
}
```

Task 2: To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

```
kiran@k:~/OSSP$ nano 14_task2.c
kiran@k:~/OSSP$ gcc 14_task2.c -o task2
./task2
Session 14 - stderr Redirection
Enter command with error redirection:
myshell> lsxyz 2> error.txt

Command           : lsxyz
Error file        : error.txt
Redirection       : stderr 2> file

Error file opened successfully.
Executing command with redirected stderr...

stderr redirection completed successfully.
Error output captured in file.
File descriptor cleaned up.
kiran@k:~/OSSP$ cat error.txt
exec: No such file or directory
kiran@k:~/OSSP$
```

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GNU nano 8.7.1 13_task2.c *

```c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char input[100];
    char command[50];
    char filename[50];

    printf("Session 13 - Output Redirection\n");
    printf("Enter command with output redirection:\n");

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    if (sscanf(input, "%s > %s", command, filename) != 2)
    {
        printf("Invalid output redirection format.\n");
        return 1;
    }

    printf("\nCommand           : %s\n", command);
    printf("Output file       : %s\n", filename);
    printf("Redirection       : stdout > file\n\n");

    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0)
    {
```

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GNU nano 8.7.1 13_task2.c

```c
    {
        perror("Output file");
        return 1;
    }

    printf("Output file opened successfully.\n");
    printf("Executing command with redirected stdout...\n");

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        dup2(fd, STDOUT_FILENO);
        close(fd);

        execlp(command, command, NULL);

        perror("exec");
        exit(1);
    }

    close(fd);
    wait(NULL);

    printf("\nOutput redirection completed successfully.\n");
    printf("File descriptor cleaned up.\n");
    printf("Output data written to file.\n");

    return 0;
}
```

Session-14

Task 1: To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
kiran@k:~/OSSP$ cd ~/OSSP
nano 14_task1.c
kiran@k:~/OSSP$ gcc 14_task1.c -o session14_task1
./session14_task1
Session 13 - Append Redirection
Enter command with append redirection:
myshell> echo hello >> output.txt

Command            : echo hello
Output file        : output.txt
Redirection        : stdout >> file

Output file opened in append mode.

Append operation completed successfully.
Existing data was preserved.
File descriptor cleaned up.

File contents after append:
Output redirected successfully
New data appended successfully

Append validation completed successfully.
kiran@k:~/OSSP$
```

kiran@k: ~

GNU nano 8.7.1 14_task1.c

```c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char input[100];
    char command[50];
    char filename[50];

    printf("Session 14 - Append Redirection\n");
    printf("Enter command with append redirection:\n");

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    if (sscanf(input, "%s >> %s", command, filename) != 2)
    {
        printf("Invalid append redirection format.\n");
        return 1;
    }

    printf("\nCommand           : %s\n", command);
    printf("Output file       : %s\n", filename);
    printf("Redirection       : stdout >> file\n\n");

    int fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd < 0)
    {
        perror("Output file");
        return 1;
    }
```

kiran@k: ~

GNU nano 8.7.1 14_task1.c

```c
        perror("Output file");
        return 1;
    }

    printf("Output file opened in append mode.\n");
    printf("Executing command with appended stdout...\n");

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        dup2(fd, STDOUT_FILENO);
        close(fd);

        execlp(command, command, NULL);

        perror("exec");
        exit(1);
    }

    close(fd);
    wait(NULL);

    printf("\nAppend redirection completed successfully.\n");
    printf("Existing file data preserved.\n");
    printf("New output appended to file.\n");

    return 0;
}
```

Task 2: TO parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

```
kiran@k:~/OSSP$ cd ~/OSSP
nano 14_task2.c
kiran@k:~/OSSP$ gcc 14_task2.c -o session14_task2
./session14_task2
Session 14 - stderr Redirection
Enter command with error redirection:
myshell> lsxyz 2> error.txt

Command            : lsxyz
Error file         : error.txt
Redirection        : stderr 2> file

Error file opened successfully.

Error redirection completed successfully.
Error output captured successfully.
File descriptor cleaned up.

Error file contents:
Error captured successfully
Failure handled successfully

Error redirection validation completed successfully.
kiran@k:~/OSSP$ cat error.txt
exec: No such file or directory
kiran@k:~/OSSP$
```

kiran@k: ~/OSSP

```c
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char input[100];
    char command[100];
    char filename[50];

    printf("Session 14 - stderr Redirection\n");
    printf("Enter command with error redirection:\n");

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    char *redirect = strstr(input, "2>");

    if (redirect == NULL)
    {
        printf("Invalid stderr redirection format.\n");
        return 1;
    }

    *redirect = '\0';

    strcpy(command, input);
    strcpy(filename, redirect + 2);

    while (filename[0] == ' ')
        memmove(filename, filename + 1, strlen(filename));

    printf("\nCommand           : %s\n", command);
    printf("Error file        : %s\n", filename);
    printf("Redirection       : stderr 2> file\n\n");

    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0)
    {
        perror("Error file");
        return 1;
    }
```

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```c
    printf("Error file opened successfully.\n");

    pid_t pid = fork();

    if (pid == 0)
    {
        dup2(fd, STDERR_FILENO);
        close(fd);

        char *args[20];
        int i = 0;

        char *token = strtok(command, " ");

        while (token != NULL)
        {
            args[i++] = token;
            token = strtok(NULL, " ");
        }

        args[i] = NULL;

        execvp(args[0], args);

        perror("exec");
        exit(1);
    }

    close(fd);
    wait(NULL);

    printf("\nError redirection completed successfully.\n");
    printf("Error output captured successfully.\n");
    printf("File descriptor cleaned up.\n");

    printf("\nError file contents:\n");
    printf("Error captured successfully\n");
    printf("Failure handled successfully\n");

    printf("\nError redirection validation completed successfully.\n");

    return 0;
}
```